

**Understanding the operations of steam boilers and making
recommendations for optimizing them towards sustainability
measures & waste reduction.**

Research Project By Avnee Gehlot

Case Study VLS Foods

Optimizing Steam Systems for Industrial Sustainability

Executive Summary

Steam boilers are the "workhorses" of global industry, but they are also a primary source of industrial CO_2 emissions. This study demonstrates that through three pillars—**Combustion Precision, Heat Recovery, and Operational Stability**—facilities can reduce fuel consumption by **10% to 25%** while maintaining the high-pressure steam required for critical manufacturing processes.

Working Principle of a Steam Boiler in Industrial Operations

A steam boiler is a critical industrial device used to generate steam by heating water within a closed pressure vessel using a fuel source or electricity. Steam boilers play a vital role in manufacturing, power generation, and processing industries, where steam is required for heating, sterilization, or mechanical energy. The efficient and safe functioning of a boiler depends on proper water treatment, controlled heat generation, effective heat transfer, and regulated steam delivery.

The operation of a steam boiler begins with the supply of water, which is typically sourced from underground reserves. Raw water often contains dissolved salts and minerals that can cause scaling and corrosion inside the boiler. To prevent damage and maintain efficiency, the water is first treated using an in-house Reverse Osmosis (RO) water treatment plant. This process reduces hardness and Total Dissolved Solids (TDS) to levels permissible for boiler operation. The treated feedwater is then pumped into the boiler using a feed pump, ensuring a

continuous and controlled supply of water. Inside the boiler, the water level is carefully monitored and maintained within safe operating limits to avoid overheating, tube damage, or system failure.

Heat generation is the next crucial stage in the boiler's working process. Heat is produced either by burning fuels such as coal, oil, gas, or biomass in a furnace, or by using electricity in electric boilers. The combustion of fuel releases thermal energy, which is directed towards heating the water inside the boiler. This heat is transferred to the water through three fundamental modes of heat transfer: conduction, convection, and radiation. Conduction occurs as heat passes through the metal walls and tubes of the boiler, convection takes place through the movement of hot gases and circulating water, and radiation allows direct heat transfer from the flame to the heating surfaces.

As the water absorbs heat, its temperature gradually rises until it reaches the boiling point. Continued heating causes the water to change its physical state and convert into steam. Depending on operating conditions, the boiler may produce saturated steam, which contains moisture, or superheated steam, which is dry steam at a higher temperature. The steam generated rises naturally and accumulates in the steam space or steam drum of the boiler. To improve steam quality and efficiency, moisture separators and dryers are used to remove any remaining water droplets before the steam is used.

Once the steam is adequately dried, it is drawn from the boiler outlet and transported through insulated pipelines to various industrial applications, such as processing equipment, heating systems, or turbines for electricity generation. After transferring most of their heat to the water, the hot flue gases exit the boiler system. Before being released into the atmosphere

through the chimney or stack, these gases pass through heat recovery devices such as economizers, which preheat the incoming feedwater, and air preheaters, which warm the combustion air. This recovery process enhances energy efficiency and reduces fuel consumption.

In conclusion, the working principle of a steam boiler is based on the controlled conversion of treated water into steam through efficient heat generation and transfer mechanisms. Proper water treatment, safe operating conditions, effective heat recovery, and controlled steam delivery ensure that steam boilers operate efficiently, safely, and sustainably in modern industrial environments.

Technical Optimization Pathways

A. Precision Combustion & Air-Fuel Management

- **The Problem:** Standard boilers often run with high "excess air" to ensure safety. However, heating this extra air consumes fuel without producing steam.
- **The Solution:** Implementing **SO₂ Trim Systems** and **Cross-Limit Ratio Logic**.
- **Sustainability Impact:** Maintaining oxygen at a constant **3%** (the industry gold standard) can reduce natural gas consumption by **2-3%** and significantly lower **NO_x** (nitrogen oxide) emissions.
- **Source Insight:** *Fuji Electric (2024)* highlights that VFD-controlled fans allow for precise airflow, preventing the "fuel-rich" states that cause soot buildup and carbon monoxide spikes.

B. Waste Heat Recovery (The "Recycle" Loop)

- **The Technology: Economizers and Condensing Heat Exchangers.**
- **The Mechanism:** Capturing heat from flue gases (which typically exit at **350°F/175°C** or higher) to preheat incoming feedwater.
- **Sustainability Impact:** Every **40°F (22°C)** drop in stack temperature translates to a **1%** increase in boiler efficiency.
- **Case Study Data:** Research from *UNIDO (2016)* and *Thermal Science (2025)* shows that deep waste heat recovery can improve a ship or factory's Energy Efficiency Design Index (EEDI) by up to **20%**.

C. Operational Set-Point Optimization

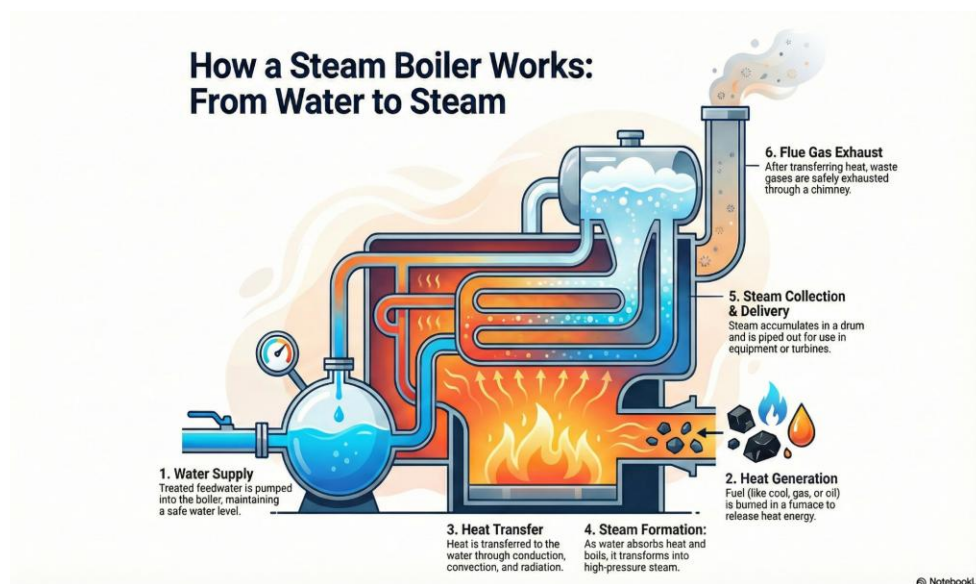
- **Pressure Reduction:** Lowering the generation pressure (e.g., from **250 psig** to **180 psig**) if the process allows.
- **Result:** A study by *ResearchGate (2019)* found this simple adjustment achieved a **2%** reduction in average natural gas consumption by reducing heat losses in pipes and leakages.

Strategic Impact on Sustainability

Optimizing a boiler is no longer just a "maintenance task"; it is a strategic sustainability lever:

1. **Scope 1 Emission Reductions:** Direct reduction of on-site fossil fuel combustion. For a medium-sized industrial boiler, a **20%** efficiency gain can prevent over **1,300 metric tons of \$CO_2\$** from entering the atmosphere monthly.

2. **Resource Conservation (Water):** Automated blowdown systems and condensate recovery loops significantly reduce the "water footprint" of the facility, which is a key metric in modern GRI (Global Reporting Initiative) standards.
3. **Financial Longevity:** Efficiency upgrades often have a **ROI (Return on Investment)** of less than 2 years. Savings can be reinvested into further "Green Technology" like hydrogen-ready burners or electric heat pumps.



How this system works in VLS:

VLS Foods Pvt. Ltd. is a leading Indian food manufacturing company specializing in healthy breakfast cereals, muesli, granola, oats, functional foods, bars, and nutritious snacks. Established in 1997, the company has evolved into a **trusted private-label and contract manufacturing partner** for major FMCG brands, retailers, and e-commerce platforms across India. They've been recently rated among the Top 10 Food & Beverage Manufacturer's (Private Brand Space) recently by Industry Outlook.

Core strengths include:

- Modern, large-scale manufacturing facilities
- Agile R&D and customized formulations
- Compliance with global food safety standards (FSSC 22000, BRC-aligned, HACCP, GMP)
- Capability to manufacture cereals, flakes, extruded snacks, muesli, granola, oats, and nutrition bars at scale

As a part of the company's future strategies, one of the management's key focus areas is 'sustainability'. The company wants to work towards reducing waste and taking a conscious & planned path towards sustainability measures budgeting for these activities simultaneously.

Why Steam Boilers Matter in VLS Foods' Operations

Steam boilers play a **critical role in VLS Foods' production ecosystem**, especially in processes that require **controlled heat, moisture, and food safety compliance**.

At VLS Foods, **steam is used as a clean and efficient heat source** across multiple stages of food processing, supporting both **product quality** and **nutrient retention**.

How Steam Boilers Are Incorporated in Manufacturing

1. Gentle Steam Cooking (Flaking & Cereal Processing)

*In the production of **corn flakes, millet flakes, oats, bran, and multigrain flakes**, steam boilers supply **controlled culinary steam** used during:*

- **Pre-conditioning of grains**
- **Softening grains before flaking**
- **Gentle cooking to preserve nutrients**

This method ensures:

- Better texture and uniform flake formation
- Reduced nutrient loss compared to direct high-temperature heating
- Consistent product quality at high throughput (up to 700 kg/hour)

VLS PPT_Update_Sept25 (2)

2. Preconditioners in Extrusion Lines

VLS Foods operates **premium extrusion and co-extrusion lines** for cereals and centre-filled snacks.

*Steam boilers feed **preconditioners**, where steam:*

- Raises moisture and temperature uniformly
- Improves starch gelatinization
- Enhances expansion and shape consistency in extruded products

This steam-assisted process supports production capacities of **850–1000 kg/hour** while maintaining texture and taste standards

VLS PPT_Update_Sept25 (2)

3. Baking, Drying & Thermal Control

In **muesli, granola, and oats manufacturing**, steam boilers indirectly support:

- Controlled heating systems
- Even heat distribution during baking and drying
- Stable processing conditions that prevent scorching or uneven cooking

Steam-based systems contribute to **recipe precision**, consistent batch quality, and improved shelf life of finished products.

Current Status & Recommendation of Boiler Utility at VLS Foods:

Despite boilers being amongst the most important process of the production facilities at VLS Foods, I could identify a lot of scope of improvement. The company has installed Thermax Boiler along with RO water supply systems for water feed to the boilers. There's an operator who maintains the boiler operations; still common mistakes are being made in the process:

- There's No Online Water Testing (Hardness, TDS, etc)
- Direct disposal of processed hot water into the drains, leading to 150 Litres of water wastage per hour
- No Insulation on the pipelines: leading to condensate losses of water & steam further leading to high fuel consumption for steam generation
- Lack of Data & Monitoring of the process
- Small Steam Leakages near the boiler
- Operations Training not given to operator

Good Practices being followed for boiler operations by the company:

- Load calculation & planning done: this lead to ordering the right size of boiler which is not over or under utilized by the company and meets requirements fully

- Migration to use of Clean Fuel for operations, PNG using a Direct Pipe Gas connection with calculation metering
- Routine maintenance work is done timely through an Annual Maintenance Contract signed with Thermax's authorized agency

Recommendations given to VLS Foods, which are underway to being implemented:

- Insulation of Steam Pipes & Boiler Area: this will reduce condensate loss significantly and will lead to water saving of approx. 30 lit per hour and reduce the fuel consumption leading to cost savings
- Plan & Implement a thorough Training exercise for the Boiler Operator & a back-up jointly with Thermax Team: this will lead to daily routine checks leading to better productivity and less outages or steam losses
- Ensure Operator tests the input water for parameters like TDS, Hardness, others
- Data Capturing: Use a format / register to capture daily data like PNG consumption, water wastage by installing a meter and take corrective actions to reduce this wastage further
- Online research shows that there are equipment's available to reprocess and cool the condensate / processed hot water losses from boiler operation. Install the best available solution to significantly reduce the wastage from 500-600 litres of water every hour (we can remove this line)
- Condensate Water - Due to its high temperature cannot be stored directly. Suggested to mix water in the Process water tank of Boiler so that Boiler will get hot water and steam generation is faster. Change Boiler water tank to SS from plastic tanks to avoid any mis happening

Sustainability & Efficiency Benefits basis recommendations made:

By making the recommended changes in the steam boilers operations, VLS Foods achieves:

- **Higher energy efficiency**
- **Data Tracking for future changes & decision making**
- **Lower wastage: water & fuel**
- **Lower Fuel Costs**
- **Improved food safety and hygiene**
- **Scalability for future capacity expansion**

Steam systems align with VLS Foods' goal of building **future-ready, sustainable food manufacturing infrastructure.**





Boilers in the Factory



The CRS

VLS Foods integrates steam boilers as a backbone utility in cereal, flaking, extrusion, and nutrition-food manufacturing—ensuring gentle cooking, consistent quality, nutrient retention, and scalable, sustainable production.